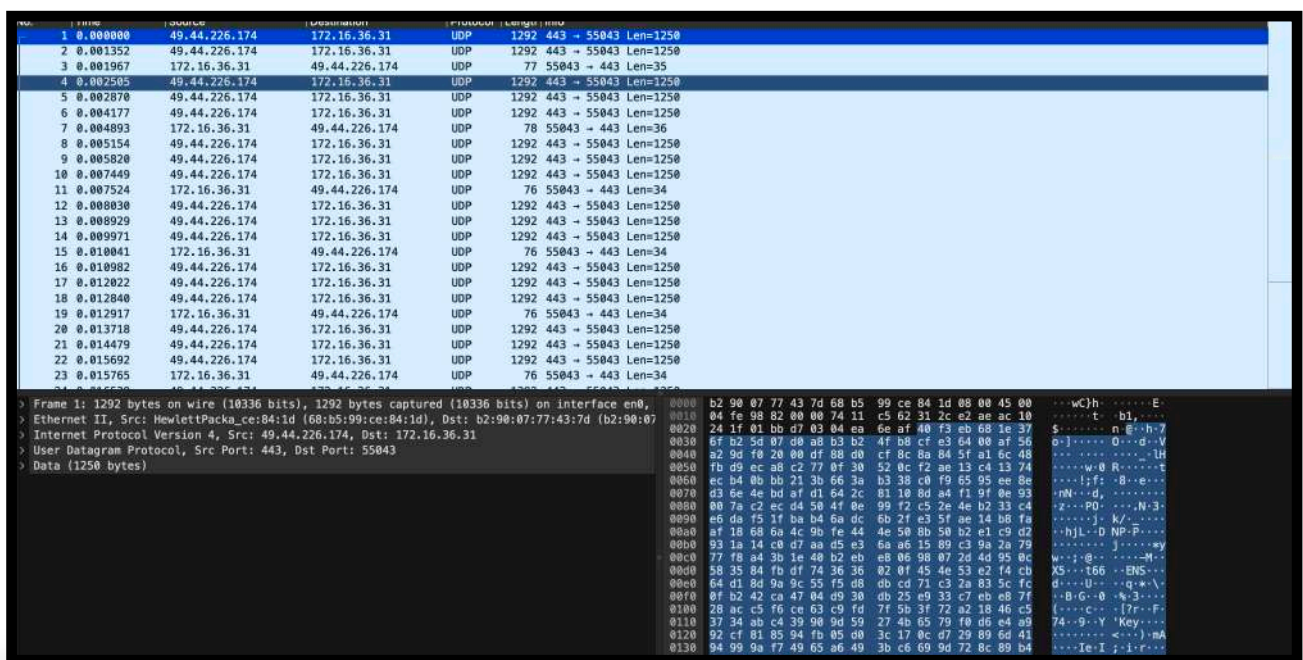


Lab Experiment — 3 : Network Analysis using Wireshark Tool

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Problem Tasks:

1. Start Wireshark and capture packets on the active network interface.



2. Filter traffic to analyze only HTTP packets and identify the websites accessed.

Using the `http` filter in Wireshark, HTTP traffic was isolated. A GET request was observed, and the **Host field** revealed that the accessed site was `neverssl.com`. This demonstrates how Wireshark can be used to identify visited websites from unencrypted HTTP requests.

Figure : HTTP GET request showing Host field (`TheProject.HTML`).

The `http` filter was applied to isolate HTTP traffic. A captured GET request to `http://home.web.cern.ch/topics/birth-web` was observed, with the **Host field** indicating `home.web.cern.ch`. This confirms that Wireshark can be used to identify websites accessed over unencrypted HTTP.

Figure X: HTTP GET request captured for `home.web.cern.ch`.

```
> Frame 26620: 558 bytes on wire (4464 bits), 558 bytes captured (4464 bits) on interface en0, id 0
> Ethernet II, Src: b2:90:07:77:43:7d (b2:90:07:77:43:7d), Dst: HewlettPacka_ce:84:1d (68:b5:99:ce:84:1d)
> Internet Protocol Version 4, Src: 172.16.36.31, Dst: 188.184.96.84
> Transmission Control Protocol, Src Port: 63283, Dst Port: 80, Seq: 1, Ack: 1, Len: 492
< Hypertext Transfer Protocol
  < GET /topics/birth-web HTTP/1.1\r\n
    Host: home.web.cern.ch\r\n
    Connection: keep-alive\r\n
    DNT: 1\r\n
    Upgrade-Insecure-Requests: 1\r\n
    User-Agent: Mozilla/5.0 (Macintosh; Intel Mac OS X 10_15_7) AppleWebKit/537.36 (KHTML, like Gecko) Chrome/140.0.0.0 Safari/537.36\r\n
    Accept: text/html,application/xhtml+xml,application/xml;q=0.9,image/avif,image/webp,image/apng,*/*;q=0.8,application/signed-exchange;v=b3;q=0.7\r\n
    Referer: http://info.cern.ch/\r\n
    Accept-Encoding: gzip, deflate\r\n
    Accept-Language: en-US,en;q=0.9\r\n
  \r\n
  [Response in frame: 26647]
  [Full request URI: http://home.web.cern.ch/topics/birth-web]
```

0000	68 b5 99 ce 84 1d b2 90 07 77 43 7d 08 00 45 00	h.....wC}..E.
0010	02 20 00 00 40 00 40 06 4b 9c ac 10 24 1f bc b8	...@.K...\$...
0020	60 54 f7 33 00 50 02 ac fa d4 fe bd 53 2e 80 18	T.3.P...S...
0030	08 0b 2f d9 00 00 01 01 08 0a 87 53 da 4d 9a d7	../. ...S.M...
0040	69 a5 47 45 54 20 2f 74 6f 70 69 63 73 2f 62 69	i-GET /t opics/bi
0050	72 74 68 2d 77 65 62 20 48 54 54 50 2f 31 2e 31	rth-web HTTP/1.1
0060	0d 0a 48 6f 73 74 3a 20 68 6f 6d 65 2e 77 65 62	..Host: home.web
0070	2e 63 65 72 6e 2e 63 68 0d 0a 43 6f 6e 6e 65 63	.cern.ch ..Connec
0080	74 69 6f 6e 3a 20 6b 65 65 70 2d 61 6c 69 76 65	tion: ke ep-alive

```
> Frame 21338: 567 bytes on wire (4536 bits), 567 bytes captured (4536 bits) on interface en0, id 0
> Ethernet II, Src: b2:90:07:77:43:7d (b2:90:07:77:43:7d), Dst: HewlettPacka_ce:84:1d (68:b5:99:ce:84:1d)
> Internet Protocol Version 4, Src: 172.16.36.31, Dst: 188.184.67.127
> Transmission Control Protocol, Src Port: 63274, Dst Port: 80, Seq: 1, Ack: 1, Len: 501
< Hypertext Transfer Protocol
  < GET /hypertext/www/TheProject.html HTTP/1.1\r\n
    Host: info.cern.ch\r\n
    Connection: keep-alive\r\n
    DNT: 1\r\n
    Upgrade-Insecure-Requests: 1\r\n
    User-Agent: Mozilla/5.0 (Macintosh; Intel Mac OS X 10_15_7) AppleWebKit/537.36 (KHTML, like Gecko) Chrome/140.0.0.0 Safari/537.36\r\n
    Accept: text/html,application/xhtml+xml,application/xml;q=0.9,image/avif,image/webp,image/apng,*/*;q=0.8,application/signed-exchange;v=b3;q=0.7\r\n
    Referer: http://info.cern.ch/\r\n
    Accept-Encoding: gzip, deflate\r\n
    Accept-Language: en-US,en;q=0.9\r\n
  \r\n
  [Response in frame: 21353]
  [Full request URI: http://info.cern.ch/hypertext/www/TheProject.html]
```

0040	c4 71 47 45 54 20 2f 68 79 70 65 72 74 65 78 74	..qGET /h ypertext
0050	2f 57 57 57 2f 54 68 65 50 72 6f 6a 65 63 74 2e	/www/The Project.
0060	68 74 6d 6c 20 48 54 54 50 2f 31 2e 31 0d 0a 48	html HTTP/1.1..H
0070	6f 73 74 3a 20 69 6e 66 6f 2e 63 65 72 6e 2e 63	ost: inf o.cern.c
0080	68 0d 0a 43 6f 6e 6e 65 63 74 69 6f 6e 3a 20 6b	h..Conne ction: k
0090	65 65 70 2d 61 6c 69 76 65 0d 0a 44 4e 54 3a 20	ee p-aliv e..DNT:
00a0	31 0d 0a 55 70 67 72 61 64 65 2d 49 6e 73 65 63	1..Upgra de-Insec
00b0	75 72 65 2d 52 65 71 75 65 73 74 73 3a 20 31 0d	ure-Requ ests: 1.
00c0	0a 55 73 65 72 2d 41 67 65 6e 74 3a 20 4d 6f 7a	..User-Ag ent: Moz

Text item (text), 45 bytes

☒ Show packet bytes Layout: Vertical (Stacked)

Wireshark · Packet 116492 · Wi-Fi

▼ Hypertext Transfer Protocol

- ▶ HEAD /edgedl/diffgen-puffin/ggkkehgbnfjpeggfpleeakpidbkibbm/0ba595cbe82f
 Connection: Keep-Alive\r\n
 Accept: */*\r\n
 Accept-Encoding: identity\r\n
 User-Agent: Microsoft BITS/7.8\r\n
 Host: edgedl.me.gvt1.com\r\n
 \r\n
 [Response in frame: 116493]
 [Full request URI: http://edgedl.me.gvt1.com/edgedl/diffgen-puffin/ggkkehgbnfjpeggfpleeakpidbkibbm/0ba595cbe82f2994dc8d8fe9111efddcd42a1778f671d633a6ec3aea1b2ea9 HTTP/1.1]

0030 00 fd 0e 56 00 00 48 45 41 44 20 2f 65 64 67 65 ...V..HE AD /edge
 0040 64 6c 2f 64 69 66 66 67 65 6e 2d 70 75 66 66 69 dl/diffg en-puffi
 0050 6e 2f 67 67 6b 6b 65 68 67 62 6e 66 6a 70 65 67 n/ggkkeh gbnfjpe
 0060 67 66 70 6c 65 65 61 6b 70 69 64 62 6b 69 62 62 gfpleeak pidbkibb
 0070 6d 6e 2f 30 62 61 35 39 35 63 62 65 38 32 38 37 mn/0ba59 5cbe8287
 0080 66 32 39 39 34 64 63 38 64 38 66 65 39 31 31 31 f2994dc8 d8fe9111
 0090 65 66 64 64 63 64 34 32 61 31 37 37 38 66 36 37 efddcd42 a1778f67
 00a0 31 64 36 33 33 61 36 65 63 33 61 65 61 31 62 32 1d633a6e c3aea1b2
 00b0 65 61 39 20 48 54 54 50 2f 31 2e 31 0d 0a 43 6f ea9 HTTP /1.1 · Co
 00c0 6e 6e 65 63 74 69 6f 6e 3a 20 4b 65 65 70 2d 41 nnection : Keep-A
 00d0 6c 69 76 65 0d 0a 41 63 63 65 70 74 3a 20 2a 2f live ·Ac cept: */
 00e0 2a 0d 0a 41 63 63 65 70 74 2d 45 6e 63 6f 64 69 * ·Accep t-Encodi
 00f0 6e 67 3a 20 69 64 65 6e 74 69 74 79 0d 0a 55 73 ng: iden tity ·Us
 0100 65 72 2d 41 67 65 6e 74 3a 20 4d 69 63 72 6f 73 er-Agent : Micros

No.: 116492 · Time: 414.031322 · Source: 172.16.88.193 · De...994dc8d8fe9111efddcd42a1778f671d633a6ec3aea1b2ea9 HTTP/1.1

✓ Show packet bytes Layout: Vertical (Stacked) Close Help

3. Use filters to locate DNS requests and identify domain names being resolved.

No.	Time	Source	Destination	Protocol	Length	Info
353...	77.093425	172.16.32.1	172.16.36.31	DNS	102	Standard query response 0xfa48 HTTPS fonts.gstatic.com
353...	77.093428	172.16.32.1	172.16.36.31	DNS	93	Standard query response 0xebf7 A fonts.gstatic.com A 142.251.222.195
370...	78.031636	172.16.36.31	172.16.32.1	DNS	74	Standard query 0x0f15 A www.google.com
370...	78.031679	172.16.36.31	172.16.32.1	DNS	74	Standard query 0x0774 HTTPS www.google.com
370...	78.036470	172.16.36.31	172.16.32.1	DNS	71	Standard query 0xaa26 A i.ytimg.com
370...	78.036511	172.16.36.31	172.16.32.1	DNS	71	Standard query 0xb0c1 HTTPS i.ytimg.com
370...	78.037078	172.16.32.1	172.16.36.31	DNS	162	Standard query response 0x0f15 A www.google.com A 142.251.221.164 NS ns4.google
370...	78.037080	172.16.32.1	172.16.36.31	DNS	171	Standard query response 0x0774 HTTPS www.google.com HTTPS NS ns2.google.com NS
370...	78.038091	172.16.32.1	172.16.36.31	DNS	247	Standard query response 0xaa26 A i.ytimg.com A 142.250.4.119 A 142.251.10.119 A
370...	78.056156	172.16.36.31	172.16.32.1	DNS	81	Standard query 0xa2f3 A jnn-pa.googleapis.com
370...	78.056197	172.16.36.31	172.16.32.1	DNS	81	Standard query 0xaf34 HTTPS jnn-pa.googleapis.com
370...	78.056558	172.16.36.31	172.16.32.1	DNS	73	Standard query 0x98f1 A yt3.ggpht.com
370...	78.056594	172.16.36.31	172.16.32.1	DNS	73	Standard query 0x3a32 HTTPS yt3.ggpht.com
371...	78.071558	172.16.32.1	172.16.36.31	DNS	134	Standard query response 0x98f1 A yt3.ggpht.com CNAME photos-ugc.l.googleusercontent
371...	78.077123	172.16.32.1	172.16.36.31	DNS	337	Standard query response 0xa2f3 A jnn-pa.googleapis.com A 142.251.223.10 A 142.2
371...	78.084654	172.16.32.1	172.16.36.31	DNS	128	Standard query response 0xb0c1 HTTPS i.ytimg.com SOA ns1.google.com
371...	78.086363	172.16.32.1	172.16.36.31	DNS	138	Standard query response 0xaf34 HTTPS jnn-pa.googleapis.com SOA ns1.google.com
371...	78.108833	172.16.32.1	172.16.36.31	DNS	175	Standard query response 0x3a32 HTTPS yt3.ggpht.com CNAME photos-ugc.l.googleusercontent
373...	78.169697	172.16.36.31	172.16.32.1	DNS	75	Standard query 0x5932 A www.gstatic.com
373...	78.169746	172.16.36.31	172.16.32.1	DNS	75	Standard query 0x8240 HTTPS www.gstatic.com
373...	78.172690	172.16.32.1	172.16.36.31	DNS	91	Standard query response 0x5932 A www.gstatic.com A 142.251.43.35
373...	78.178189	172.16.32.1	172.16.36.31	DNS	132	Standard query response 0x8240 HTTPS www.gstatic.com SOA ns1.google.com
426...	82.704807	172.16.36.31	172.16.32.1	DNS	75	Standard query 0x973e A play.google.com
426...	82.704878	172.16.36.31	172.16.32.1	DNS	75	Standard query 0xce8b HTTPS play.google.com
426...	82.707094	172.16.32.1	172.16.36.31	DNS	163	Standard query response 0x973e A play.google.com A 142.251.222.174 NS ns1.google
426...	82.712423	172.16.36.31	172.16.36.31	DNS	125	Standard query response 0xce8b HTTPS play.google.com SOA ns1.google.com
440...	85.010233	172.16.36.31	172.16.32.1	DNS	85	Standard query 0xc3d5 A gateway.fe2.apple-dns.net

```

> Frame 86957: 77 bytes on wire (616 bits), 77 bytes captured (616 bits) on interface en0, id 0
> Ethernet II, Src: b2:90:07:77:43:7d (b2:90:07:77:43:7d), Dst: HewlettPacka_ce:84:1d (68:b5:99:ce:84:1d)
> Internet Protocol Version 4, Src: 172.16.36.31, Dst: 172.16.32.1
< User Datagram Protocol, Src Port: 37597, Dst Port: 53
  Source Port: 37597
  Destination Port: 53
  Length: 43
  Checksum: 0x7eea [unverified]
  [Checksum Status: Unverified]
  [Stream index: 246]
  [Stream Packet Number: 1]
  > [Timestamps]
  UDP payload (35 bytes)
< Domain Name System (query)
  Transaction ID: 0xb27b
  > Flags: 0x0100 Standard query
  Questions: 1
  Answer RRs: 0
  Authority RRs: 0
  Additional RRs: 0
  > Queries
    [Response In: 87428]

0000  68 b5 99 ce 84 1d b2 90 07 77 43 7d 08 00 45 00  h.....wC}..E.
0010  00 3f 5f 57 00 00 40 11 7f 16 ac 10 24 1f ac 10  .?_W..@. ....$.
0020  20 01 02 dd 00 35 00 2b 7e ea b2 7b 01 00 00 01  ....5.+...{....
0030  00 00 00 00 00 00 03 77 77 77 09 77 69 6b 69 70  ....w ww-wikip
0040  65 64 69 61 03 6f 72 67 00 00 01 00 01          edia.org .....

```

```

> Frame 19127: 74 bytes on wire (592 bits), 74 bytes captured (592 bits) on interface en0, id 0
> Ethernet II, Src: b2:90:07:77:43:7d (b2:90:07:77:43:7d), Dst: HewlettPacka_ce:84:1d (68:b5:99:ce:84:1d)
> Internet Protocol Version 4, Src: 172.16.36.31, Dst: 172.16.32.1
< User Datagram Protocol, Src Port: 32942, Dst Port: 53
  Source Port: 32942
  Destination Port: 53
  Length: 40
  Checksum: 0x3999 [unverified]
  [Checksum Status: Unverified]
  [Stream index: 55]
  [Stream Packet Number: 1]
  > [Timestamps]
  UDP payload (32 bytes)
< Domain Name System (query)
  Transaction ID: 0xa1a39
  > Flags: 0x0100 Standard query
  Questions: 1
  Answer RRs: 0
  Authority RRs: 0
  Additional RRs: 0
  > Queries
    [Response In: 19153]

0000  68 b5 99 ce 84 1d b2 90 07 77 43 7d 08 00 45 00  h.....wC}..E.
0010  00 3c a1 aa 00 00 40 11 3c c6 ac 10 24 1f ac 10  .<....@. <...$.
0020  20 01 80 ae 00 35 00 28 39 99 1a 39 01 00 00 01  ....5.( 9..9....
0030  00 00 00 00 00 00 03 77 77 77 06 67 6f 6f 67 6c  ....w ww-googl
0040  65 03 63 6f 6d 00 00 01 00 01          e.com... ..

```

4. Analyze TCP 3-way handshake packets to confirm successful connection establishment.

tcp.flags.syn == 1(filter)

tcp.port == 80 || udp.port == 80

A complete TCP 3-way handshake was captured between the client (172.16.36.31) and the server (188.184.67.127).

- The client initiated with a **SYN** packet.
- The server replied with **SYN, ACK**.
- The client then sent an **ACK**, completing the handshake.

This confirms that a TCP connection was successfully established.

Figure : Wireshark capture showing SYN → SYN/ACK → ACK handshake sequence

178..	51.441973	172.16.36.31	34.223.124.45	TCP	78	[TCP Retransmission] 63265 → 80 [SYN] Seq=0 Win=65535 Len=0 MSS=1460 WS=64 TSval=3722882718 TSecr=0
212..	57.983094	172.16.36.31	188.184.67.127	TCP	78	63274 → 80 [SYN] Seq=0 Win=65535 Len=0 MSS=1460 WS=64 TSval=2219631217 TSecr=0 SACK_PERM
213..	58.226431	188.184.67.127	172.16.36.31	TCP	74	80 → 63274 [SYN, ACK] Seq=0 Ack=1 Win=31856 Len=0 MSS=1460 SACK_PERM TSval=3370370161 TSecr=2219631217
213..	58.226627	172.16.36.31	188.184.67.127	TCP	66	63274 → 80 [ACK] Seq=1 Ack=1 Win=131776 Len=0 TSval=2219631461 TSecr=3370370161
213..	58.226956	172.16.36.31	188.184.67.127	HTTP	567	GET /hypertext/WWW/TheProject.html HTTP/1.1
213..	58.471381	188.184.67.127	172.16.36.31	TCP	66	80 → 63274 [ACK] Seq=1 Ack=502 Win=31872 Len=0 TSval=3370370406 TSecr=2219631461
213..	58.472565	188.184.67.127	172.16.36.31	TCP	1514	80 → 63274 [ACK] Seq=1 Ack=502 Win=31872 Len=1448 TSval=3370370406 TSecr=2219631461 [TCP PDU reassem
213..	58.472569	188.184.67.127	172.16.36.31	HTTP	1068	HTTP/1.1 200 OK (text/html)
213..	58.472690	172.16.36.31	188.184.67.127	TCP	66	63274 → 80 [ACK] Seq=502 Ack=2451 Win=129344 Len=0 TSval=2219631707 TSecr=3370370406
213..	58.472754	188.184.67.127	172.16.36.31	TCP	66	80 → 63274 [FIN, ACK] Seq=2451 Ack=502 Win=31872 Len=0 TSval=3370370406 TSecr=2219631461
213..	58.472829	172.16.36.31	188.184.67.127	TCP	66	63274 → 80 [ACK] Seq=502 Ack=2452 Win=129344 Len=0 TSval=2219631707 TSecr=3370370406
213..	58.473939	172.16.36.31	188.184.67.127	TCP	66	63274 → 80 [FIN, ACK] Seq=502 Ack=2452 Win=131072 Len=0 TSval=2219631708 TSecr=3370370406
213..	58.716527	188.184.67.127	172.16.36.31	TCP	66	80 → 63274 [ACK] Seq=2452 Ack=503 Win=31872 Len=0 TSval=3370370651 TSecr=2219631708
238..	64.803492	172.16.36.31	34.223.124.45	TCP	54	[TCP Keep-Alive] 63263 → 80 [ACK] Seq=0 Ack=1 Win=131712 Len=0
242..	65.054572	34.223.124.45	172.16.36.31	TCP	66	[TCP Window Update] 80 → 63263 [ACK] Seq=1 Ack=1 Win=29312 Len=0 TSval=3425826701 TSecr=220274924
265..	67.007269	172.16.36.31	188.184.96.84	TCP	78	63283 → 80 [SYN] Seq=0 Win=65535 Len=0 MSS=1460 WS=64 TSval=2270419293 TSecr=0 SACK_PERM
266..	67.190705	172.16.36.31	188.184.96.84	TCP	78	63284 → 80 [SYN] Seq=0 Win=65535 Len=0 MSS=1460 WS=64 TSval=315075864 TSecr=0 SACK_PERM
266..	67.247442	188.184.96.84	172.16.36.31	TCP	74	80 → 63283 [SYN, ACK] Seq=0 Ack=1 Win=65160 Len=0 MSS=1460 SACK_PERM TSval=2597808549 TSecr=22704192
266..	67.247551	172.16.36.31	188.184.96.84	TCP	66	63283 → 80 [ACK] Seq=1 Ack=1 Win=131776 Len=0 TSval=2270419533 TSecr=2597808549
266..	67.247660	172.16.36.31	188.184.96.84	HTTP	558	GET /topics/birth-web HTTP/1.1
266..	67.426254	188.184.96.84	172.16.36.31	TCP	74	80 → 63284 [SYN, ACK] Seq=0 Ack=1 Win=65160 Len=0 MSS=1460 SACK_PERM TSval=2597808735 TSecr=31507586
266..	67.426399	172.16.36.31	188.184.96.84	TCP	66	63284 → 80 [ACK] Seq=1 Ack=1 Win=131776 Len=0 TSval=315076100 TSecr=2597808735
266..	67.489458	188.184.96.84	172.16.36.31	HTTP	185	HTTP/1.1 302 Found
266..	67.489563	172.16.36.31	188.184.96.84	TCP	66	63283 → 80 [ACK] Seq=493 Ack=120 Win=131712 Len=0 TSval=2270419775 TSecr=2597808791

A common example where we can see many hand shakes

```
78 63274 → 80 [SYN] Seq=0 Win=65535 Len=0 MSS=1460 WS=64 TSval=2270419293 TSecr=0 SACK_PERM
74 80 → 63274 [SYN, ACK] Seq=0 Ack=1 Win=31856 Len=0 MSS=1460 SACK_PERM TSval=3370370161 TSecr=2219631217
66 63274 → 80 [ACK] Seq=1 Ack=1 Win=131776 Len=0 TSval=2219631461 TSecr=3370370161
567 GET /hypertext/WWW/TheProject.html HTTP/1.1
66 80 → 63274 [ACK] Seq=1 Ack=502 Win=31872 Len=0 TSval=3370370406 TSecr=2219631461
1514 80 → 63274 [ACK] Seq=1 Ack=502 Win=31872 Len=1448 TSval=3370370406 TSecr=2219631461 [TCP PDU reassem
1068 HTTP/1.1 200 OK (text/html)
66 63274 → 80 [ACK] Seq=502 Ack=2451 Win=129344 Len=0 TSval=2219631707 TSecr=3370370406
66 80 → 63274 [FIN, ACK] Seq=2451 Ack=502 Win=31872 Len=0 TSval=3370370406 TSecr=2219631461
66 63274 → 80 [ACK] Seq=502 Ack=2452 Win=129344 Len=0 TSval=2219631707 TSecr=3370370406
66 63274 → 80 [FIN, ACK] Seq=502 Ack=2452 Win=131072 Len=0 TSval=2219631708 TSecr=3370370406
66 80 → 63274 [ACK] Seq=2452 Ack=503 Win=31872 Len=0 TSval=3370370651 TSecr=2219631708
54 [TCP Keep-Alive] 63263 → 80 [ACK] Seq=0 Ack=1 Win=131712 Len=0
66 [TCP Window Update] 80 → 63263 [ACK] Seq=1 Ack=1 Win=29312 Len=0 TSval=3425826701 TSecr=220274924
```

```
78 63284 → 80 [SYN] Seq=0 Win=65535 Len=0 MSS=1460 WS=64 TSval=2270419293 TSecr=0 SACK_PERM
74 80 → 63283 [SYN, ACK] Seq=0 Ack=1 Win=65160 Len=0 MSS=1460 SACK_PERM TSval=2597808549 TSecr=22704192
66 63283 → 80 [ACK] Seq=1 Ack=1 Win=131776 Len=0 TSval=2270419533 TSecr=2597808549
558 GET /topics/birth-web HTTP/1.1
74 80 → 63284 [SYN, ACK] Seq=0 Ack=1 Win=65160 Len=0 MSS=1460 SACK_PERM TSval=2597808735 TSecr=31507586
66 63284 → 80 [ACK] Seq=1 Ack=1 Win=131776 Len=0 TSval=315076100 TSecr=2597808735
185 HTTP/1.1 302 Found
66 63283 → 80 [ACK] Seq=493 Ack=120 Win=131712 Len=0 TSval=2270419775 TSecr=2597808791
```

5. Detect any suspicious activity, such as repeated connections from unknown IP addresses.

During the capture, repeated connection attempts were observed from the client 172.16.36.31 to the server 17.248.239.65 on port 443. Wireshark flagged these as **TCP retransmissions of SYN packets**, meaning the server did not respond with a SYN/ACK.

This behavior indicates either network issues, firewall blocking, or potentially suspicious scanning activity. Repeated failed handshakes to unknown external IP addresses raise a security concern

813...	76.187032	49.44.226.174	172.16.36.31	UDP	1292	443 → 55043	Len=1250	
813...	76.188566	49.44.226.174	172.16.36.31	UDP	1292	443 → 55043	Len=1250	
813...	76.189337	49.44.226.174	172.16.36.31	UDP	1292	443 → 55043	Len=1250	
813...	76.189402	172.16.36.31	49.44.226.174	UDP	81	55043 → 443	Len=39	
813...	76.190202	49.44.226.174	172.16.36.31	UDP	1292	443 → 55043	Len=1250	
813...	76.190838	172.16.36.31	17.248.239.65	TCP	78	[TCP Retransmission] 64390 → 443 [SYN] Seq=0 Win=0 Len=0 MSS=1460 WS=64 TSval=1144624443 TSecr=0 SACK_PERM		
813...	76.190850	172.16.36.31	142.250.205.42	TCP	78	[TCP Retransmission] 64386 → 443 [SYN] Seq=0 Win=0 Len=0 MSS=1460 WS=64 TSval=3575418339 TSecr=0 SACK_PERM		
813...	76.190887	172.16.36.31	17.248.239.65	TCP	78	[TCP Retransmission] 64389 → 443 [SYN] Seq=0 Win=0 Len=0 MSS=1460 WS=64 TSval=3553247160 TSecr=0 SACK_PERM		
813...	76.191645	49.44.226.174	172.16.36.31	UDP	1292	443 → 55043	Len=1250	
813...	76.192519	172.16.36.31	49.44.226.174	UDP	77	55043 → 443	Len=35	
813...	76.192703	49.44.226.174	172.16.36.31	UDP	1292	443 → 55043	Len=1250	
813...	76.192766	172.16.36.31	49.44.226.174	UDP	78	55043 → 443	Len=36	
813...	76.193638	49.44.226.174	172.16.36.31	UDP	1292	443 → 55043	Len=1250	
813...	76.193734	172.16.36.31	49.44.226.174	UDP	79	55043 → 443	Len=37	
813...	76.194597	49.44.226.174	172.16.36.31	UDP	1292	443 → 55043	Len=1250	
813...	76.194667	172.16.36.31	49.44.226.174	UDP	77	55043 → 443	Len=35	
813...	76.195461	49.44.226.174	172.16.36.31	UDP	1292	443 → 55043	Len=1250	
813...	76.196148	49.44.226.174	172.16.36.31	UDP	1292	443 → 55043	Len=1250	
813...	76.196240	172.16.36.31	49.44.226.174	UDP	78	55043 → 443	Len=36	

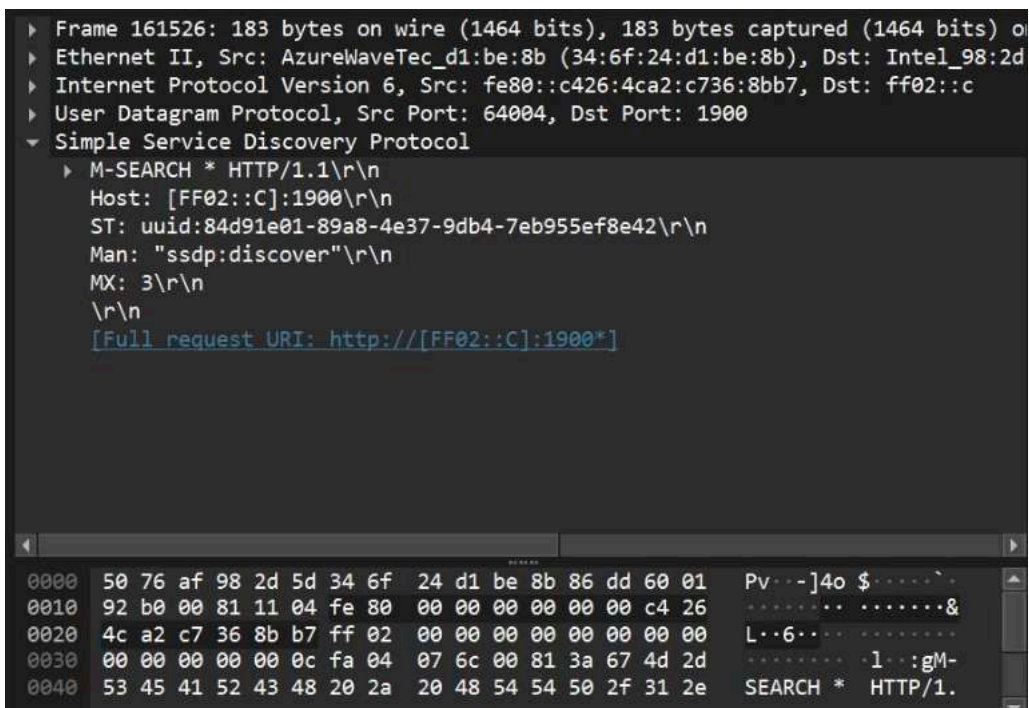
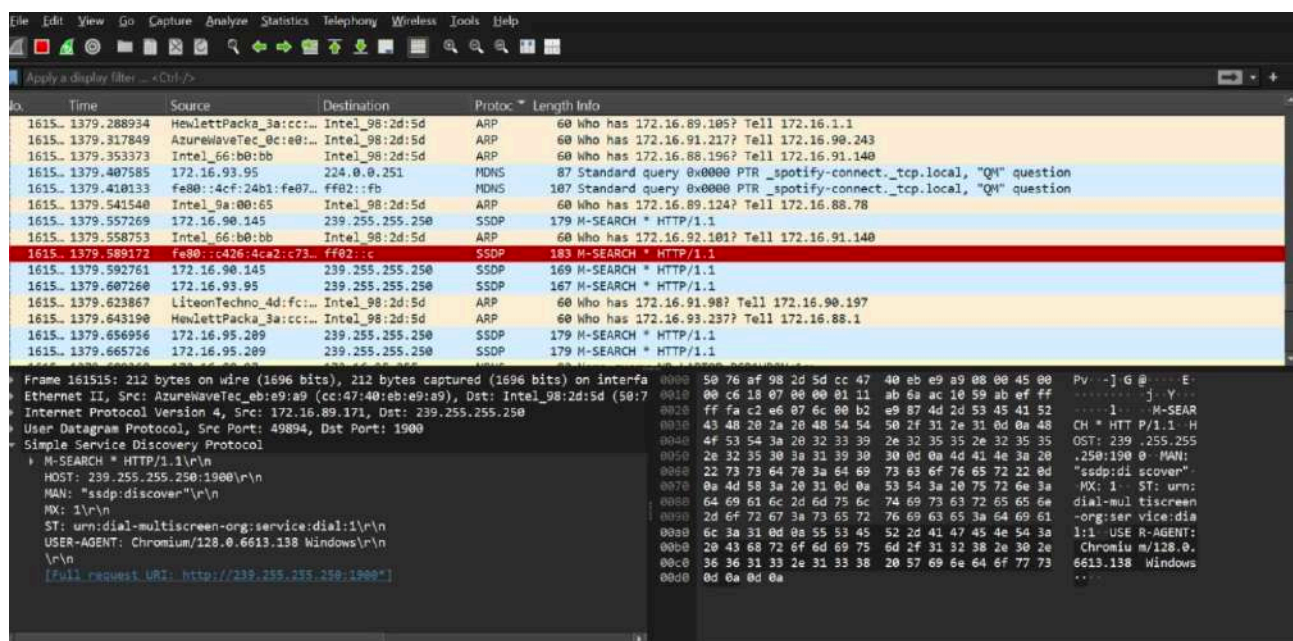
```
Transmission Control Protocol, Src Port: 64391, Dst Port: 443, Seq: 0, Len: 0
  Source Port: 64391
  Destination Port: 443
  [Stream index: 76]
  [Stream Packet Number: 1]
  > [Conversation completeness: Incomplete, SYN_SENT (1)]
  [TCP Segment Len: 0]
  Sequence Number: 0 (relative sequence number)
  Sequence Number (raw): 1602862179
  [Next Sequence Number: 1 (relative sequence number)]
  Acknowledgment Number: 0
  Acknowledgment number (raw): 0
  1011 .... = Header Length: 44 bytes (11)
  > Flags: 0x002 (SYN)
  Window: 0
  [Calculated window size: 0]
  Checksum: 0xb920 [unverified]
  [Checksum Status: Unverified]
  Urgent Pointer: 0
  > Options: (24 bytes), Maximum segment size, No-Operation (NOP), Window scale, No-Operation (NOP), No-Operation (NOP), Timestamps, SACK permiss...
  > [Timestamps]
0000  68 b5 99 ce 84 1d b2 90 07 77 43 7d 08 00 45 00  h.....wC)...E
0010  00 40 00 00 00 40 00 40 06 69 4e ac 10 24 1f 11 f8  @...@iN...$...
0020  ef 42 fb 87 01 bb 5f 89 bc 63 00 00 00 00 b0 02  B..._...c.....
0030  00 00 b9 20 00 00 02 04 05 b4 01 03 03 06 01 01  ....
0040  08 0a 52 61 40 e0 00 00 00 00 04 02 00 00  Ra@.....
```

6.Document your findings with screenshots of relevant

packet captures.

During the packet capture, I noticed red-highlighted packets in Wireshark, which typically indicate reset, errors, or multicast traffic. On further inspection, one such packet was identified as SSDP (Simple Service Discovery Protocol) traffic.

SSDP packet:

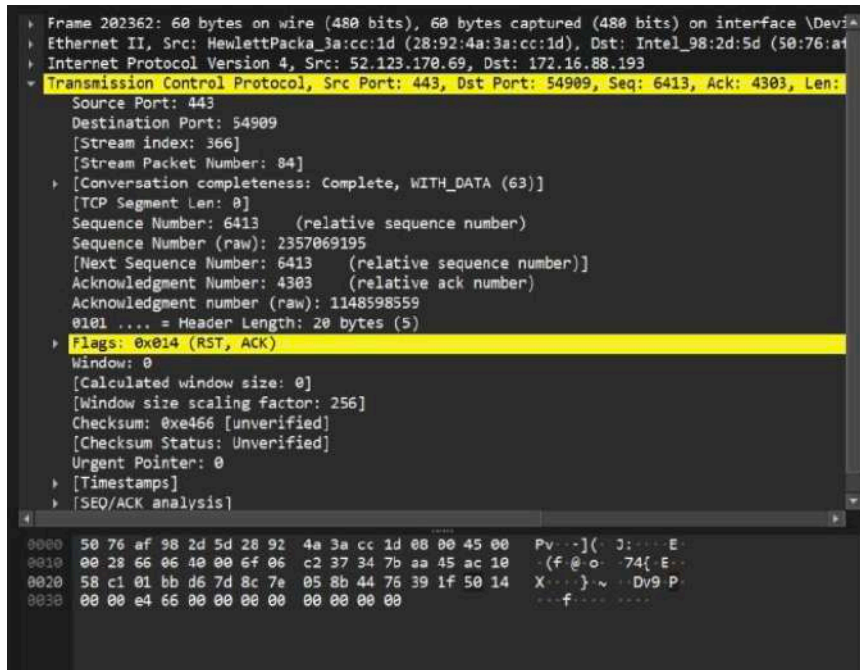


TCP packet:

Another red-highlighted packet captured was a TCP packet with the RST

(Reset) flag set. TCP RST packets are used when one side wants to abruptly

terminate a connection



Expected Outcome:

- A detailed analysis of captured traffic.
- Identification of network protocols and communication patterns.
- Detection of at least one anomaly or unusual activity.
- A final report that could be used as forensic evidence or for network troubleshooting.

All the packets captured on our college lab WiFi have been analyzed:

- We select the enp1s0 stream to capture the packets of our internet Ethernet connection of the Lab.
- Only HTTP Protocol Packets filtered out by putting the display filter http and we can see a lot of GET API requests.
- DNS query is resolving the names into IP addresses for our visited websites like www.google.com, www.vtop.ac.in, www.neopat.com
- TCP 3 way handshake found for one of the websites (neopat) while first connecting to the website.

A few IP addresses have been found suspicious for being unknown and consuming higher no of bytes than usual.